

REPLIT PROMPT — YachtWorth: Survey Report Builder

CONTEXT

Build a new tool "Survey Report Builder" inside the existing YachtWorth app. Entry point: Tools tab → Survey Report Builder (currently SOON → LIVE after build). Target user: professional marine surveyors (Surveyor plan \$89/mo). This tool generates a professional pre-purchase survey report PDF matching industry standards (YDSA / IIMS / SCMS).

OVERVIEW

Surveyor fills in a structured checklist of 26 sections on mobile. Each section has numbered items with status, notes and photos. Recommendations are auto-collected into a summary. Final output: professional multi-page PDF report.

SUPABASE SCHEMA

```
-- Survey report header
create table if not exists survey_reports (
  id uuid primary key default gen_random_uuid(),
  user_id text not null,
  yacht_id uuid references yachts(id) on delete set null,

  -- Cover info
  vessel_name text not null,
  vessel_type text,
  manufacturer text,
  model text,
  year_built integer,
  flag text,
  hin text,                      -- Hull Identification Number
  lying text,                   -- Location where surveyed
  survey_date date,
  survey_purpose text default 'Pre-purchase',
  weather_conditions text,
```

```

-- Client info
client_name text,
client_email text,

-- Surveyor info
surveyor_name text,
surveyor_qualification text, -- Dip.MS, YDSA, IIMS etc.
surveyor_company text,
surveyor_phone text,
surveyor_email text,
surveyor_logo_url text,
surveyor_signature_url text, -- drawn signature saved as image

-- Status
status text default 'draft', -- draft | complete
overall_condition text,      -- Excellent | Good | Fair | Poor

-- Counters (auto-calculated)
total_recommendations_a integer default 0,
total_recommendations_b integer default 0,
total_recommendations_c integer default 0,
total_recommendations_d integer default 0,

created_at timestampz default now(),
updated_at timestampz default now()
);

-- Individual section items
create table if not exists survey_items (
  id uuid primary key default gen_random_uuid(),
  report_id uuid references survey_reports(id) on delete cascade,
  section_number integer not null, -- 1-26
  section_name text not null,
  item_number text not null,      -- e.g. "6.0", "6.1", "15.3"
  description text,              -- what was inspected
  condition text,                -- Excellent | Serviceable | Fair | Poor |
  notes text,                    -- surveyor observations
  recommendation_level text,     -- A | B | C | D | null
  recommendation_text text,     -- full recommendation text
  photo_urls jsonb default '[]', -- array of photo URLs
  moisture_reading numeric,      -- for hull/deck items
  moisture_level text,          -- Low | Medium | High
  sort_order integer,
  created_at timestampz default now()
);

```

```

-- Sea trial data
create table if not exists survey_sea_trial (
  id uuid primary key default gen_random_uuid(),
  report_id uuid references survey_reports(id) on delete cascade,
  trial_date date,
  location text,
  weather text,
  sea_state text,
  narrative text,
  -- RPM table (JSON array of {rpm, coolant_p, coolant_s, oil_p, oil_s, speed})
  rpm_table jsonb default '[]',
  tickover_rpm integer,
  tickover_speed numeric,
  max_rpm integer,
  max_speed numeric,
  created_at timestamptz default now()
);

create index survey_reports_user_id_idx on survey_reports(user_id);
create index survey_items_report_id_idx on survey_items(report_id);

```

NAVIGATION

Tools tab

- SurveyBuilderScreen (list of saved reports + new)
 - SurveySetupScreen (cover info, vessel, client, surveyor)
 - SurveySectionsScreen (26 sections list)
 - SurveySectionScreen (individual section checklist)
 - SurveyItemScreen (individual item detail)
 - SeaTrialScreen
 - SurveyPreviewScreen (PDF preview)
 - PDF export + share
-

SCREEN 1: SURVEY LIST

SURVEY REPORTS [+ New Survey]

DRAFT

T9 · Terranova T85 · 28 Nov 2025

Antibes · Pre-purchase · 14 sections complete →

COMPLETE

Aurelia · Azimut 24M · 15 Mar 2026

Monaco · Pre-purchase · PDF generated →

Swipe left → Delete (with confirmation)

SCREEN 2: SURVEY SETUP

Four sub-sections in a scrollable form:

VESSEL

VESSEL NAME *

[T9_____]

TYPE

[Motor Yacht] [Sailing] [Catamaran] [Superyacht]

MANUFACTURER

[Terranova Yachts_____]

MODEL

[T85 M/Y_____]

YEAR OF BUILD

[2019]

FLAG / REGISTRATION

[Malta_____]

HIN (Hull Identification Number)

[IT-TRHT8509L922_____]

LYING (Current location)

[Port Vauban, Antibes_____]

DATE OF SURVEY

[ 17 28 Nov 2025]

PURPOSE

[Pre-purchase] [Insurance] [Annual] [Damage] [Other]

CLIENT

CLIENT NAME

[Vasilis Moundreas_____]

CLIENT EMAIL

[_____]

CLIENT PHONE

[_____]

SURVEYOR

YOUR NAME *

[Captain Henry Topham_____]

QUALIFICATION

[Dip.MS_____] e.g. YDSA, IIMS, Dip.MS

COMPANY

[Marine Developments Int.]

PHONE

[+34 664 890 740_____]

EMAIL

[_____]

LOGO

[ Upload logo]

SIGNATURE

[ Draw signature] ← finger signature pad

CONDITIONS

WEATHER CONDITIONS

[Sunny and calm, wind 5kts]

SEA STATE

[Calm]

SCREEN 3: SECTIONS LIST

Shows all 26 sections with completion status:

SURVEY SECTIONS

✓	1. Introduction		→
✓	2. Specification		→
	3. Glossary of Terms		→ (auto-filled)
	4. Limitations of Survey		→
✓	5. Construction		→
⚠	6. Hull	A	→ (has recommendation A)
✓	7. Rudders and Steering		→
⚠	8. Stern Gear	A	→
...			
	26. Sea Trial		→

PROGRESS: 8 of 26 sections complete

[Generate PDF Report] ← gold button, active even if incomplete

Status icons:

- ✓ green = all items filled
- ⚠ amber = has recommendations
-  = has recommendation A (urgent)
- Grey = not started

SCREEN 4: SECTION DETAIL

Example: Section 6 — Hull

← back 6. HULL

[+ Add item] [Section notes]

6.0 Hull inspection
Condition: [Not inspected ▼]
Notes: [The vessel was not hauled...]
Recommendation: [None ▼]
Photos: [ Add]
Moisture: [____] % Level: [Low] [Med] [High]

6.1 Hull condition
Condition: [Fair ▼]
Notes: [Some marine growth present...]
Recommendation: [B ▼]
Rec. text: [Lift the vessel and pressure wash...]
Photos: [] [pic055] [pic056]

[+ Add item to this section]

Condition picker (bottom sheet):

SELECT CONDITION

Excellent condition
Serviceable / Functioning
Fair condition
Poor condition
Not applicable (N/A)
Not inspected

Recommendation picker:

RECOMMENDATION LEVEL

None
A – Urgent: ensure safe operation
B – Soon: continued successful operation
C – Expert/specialist required
D – Cosmetic / leisure equipment

When level selected → text field appears for full recommendation text.

SECTIONS CONTENT — PRE-FILLED TEMPLATES

Each section has default items pre-loaded. Surveyor can edit, delete or add items.

```
const SECTION_TEMPLATES = [  
  {  
    number: 1,
```

```

    name: 'Introduction',
    items: [
      { number: '1.0', description: 'Client and location', notes: '' },
      { number: '1.1', description: 'Purpose and scope', notes: '' },
    ]
  },
  {
    number: 2,
    name: 'Specification',
    items: [
      { number: '2.0', description: 'Vessel specifications from ships papers', no
    ]
    // Auto-filled from yacht profile if linked
  },
  {
    number: 3,
    name: 'Glossary of Terms',
    isStatic: true,
    staticContent: `Excellent condition – The item described is in 'as new' condi
Serviceable / Functioning – The item described is serviceable and fit for purpose
Fair condition – The item described is serviceable but showing signs of wear.
Poor condition – The item described is in need of replacement or repair.
Moisture readings – Low, Medium or High as measured by moisture meter.
Recommendations: A – Urgent. B – Soonest opportunity. C – Expert required. D – Co
  },
  {
    number: 4,
    name: 'Limitations of Survey',
    items: [
      { number: '4.0', description: 'No parts dismantled', notes: '' },
      { number: '4.1', description: 'Covered/obscured areas', notes: '' },
      { number: '4.2', description: 'Owner declarations', notes: '' },
      { number: '4.3', description: 'Conditions', notes: '' },
    ]
  },
  {
    number: 5,
    name: 'Construction',
    items: [
      { number: '5.0', description: 'Hull construction and material', notes: '' }
      { number: '5.1', description: 'Deck construction', notes: '' },
    ]
  },
  {
    number: 6,
    name: 'Hull',
    items: [

```



```

    { number: '6.0', description: 'Hull inspection (above/below waterline)', no
    { number: '6.1', description: 'Hull condition and antifouling', notes: '' }
    { number: '6.2', description: 'Osmosis / moisture readings', notes: '' },
  ]
},
{
  number: 7,
  name: 'Rudders and Steering',
  items: [
    { number: '7.0', description: 'Rudder condition and bearings', notes: '' },
    { number: '7.1', description: 'Steering system', notes: '' },
    { number: '7.2', description: 'Autopilot', notes: '' },
  ]
},
{
  number: 8,
  name: 'Stern Gear',
  items: [
    { number: '8.0', description: 'Stern gear inspection', notes: '' },
    { number: '8.1', description: 'Shaft logs and seals (PSS)', notes: '' },
    { number: '8.2', description: 'Propellers and anodes', notes: '' },
  ]
},
{
  number: 9,
  name: 'Through Hull Fittings and Seacocks',
  items: [
    { number: '9.0', description: 'Engine raw water intakes', notes: '' },
    { number: '9.1', description: 'AC raw water intake', notes: '' },
    { number: '9.2', description: 'Generator raw water intakes', notes: '' },
    { number: '9.3', description: 'Watermaker intake', notes: '' },
    { number: '9.4', description: 'Overboard discharge seacocks', notes: '' },
  ]
},
{
  number: 10,
  name: 'Hull Topsides',
  items: [
    { number: '10.0', description: 'Topsides condition', notes: '' },
    { number: '10.1', description: 'Gelcoat / paint condition', notes: '' },
    { number: '10.2', description: 'Handrails and fendering', notes: '' },
  ]
},
{
  number: 11,
  name: 'Deck and Fittings',
  items: [

```

```

    { number: '11.0', description: 'Deck surface and overlay', notes: '' },
    { number: '11.1', description: 'Foredeck', notes: '' },
    { number: '11.2', description: 'Guard rails and stanchions', notes: '' },
    { number: '11.3', description: 'Anchor windlass / capstans', notes: '' },
    { number: '11.4', description: 'Aft capstans / winches', notes: '' },
    { number: '11.5', description: 'Wheelhouse windscreen and wipers', notes: '' },
    { number: '11.6', description: 'Fairleads and cleats', notes: '' },
    { number: '11.7', description: 'Fenders and mooring lines', notes: '' },
    { number: '11.8', description: 'Tender crane / deck crane', notes: '' },
    { number: '11.9', description: 'Passerelle / gangway', notes: '' },
    { number: '11.10', description: 'Flybridge', notes: '' },
    { number: '11.11', description: 'Anchors and chain', notes: '' },
  ]
},
{
  number: 12,
  name: 'Hatches and Port Lights',
  items: [
    { number: '12.0', description: 'Deck hatches', notes: '' },
    { number: '12.1', description: 'Cabin portlights', notes: '' },
    { number: '12.2', description: 'Saloon windows', notes: '' },
  ]
},
{
  number: 13,
  name: 'Ventilation',
  items: [
    { number: '13.0', description: 'Air conditioning system', notes: '' },
    { number: '13.1', description: 'Natural ventilation', notes: '' },
  ]
},
{
  number: 14,
  name: 'Canvas and Covers',
  items: [
    { number: '14.0', description: 'Saloon seating and cushions', notes: '' },
    { number: '14.1', description: 'Cabin mattresses and covers', notes: '' },
    { number: '14.2', description: 'Bimini / cockpit covers', notes: '' },
  ]
},
{
  number: 15,
  name: 'Engines and Transmissions',
  items: [
    { number: '15.0', description: 'Engine condition and installation', notes: '' },
    { number: '15.1', description: 'Engine mountings', notes: '' },
    { number: '15.2', description: 'Exhaust system', notes: '' },
  ]
}

```

```

    { number: '15.3', description: 'Air filters and turbochargers', notes: '' }
    { number: '15.4', description: 'Gearboxes / transmissions', notes: '' },
    { number: '15.5', description: 'Fuel system and filters', notes: '' },
    { number: '15.6', description: 'Fuel tanks', notes: '' },
    { number: '15.7', description: 'Day tanks', notes: '' },
  ]
},
{
  number: 16,
  name: 'Internal Structure and Bilges',
  items: [
    { number: '16.0', description: 'Bulkheads and structure', notes: '' },
    { number: '16.1', description: 'Stem and keel', notes: '' },
    { number: '16.2', description: 'Frames and stringers', notes: '' },
    { number: '16.3', description: 'Bilge pumps and float switches', notes: '' },
    { number: '16.4', description: 'Bilge condition', notes: '' },
  ]
},
{
  number: 17,
  name: 'Accommodation and Domestic',
  items: [
    { number: '17.0', description: 'Forward cabin(s)', notes: '' },
    { number: '17.1', description: 'Guest cabin(s)', notes: '' },
    { number: '17.2', description: 'Master cabin', notes: '' },
    { number: '17.3', description: 'Galley and appliances', notes: '' },
    { number: '17.4', description: 'Saloon and dining', notes: '' },
    { number: '17.5', description: 'Laundry', notes: '' },
    { number: '17.6', description: 'Interior finishes', notes: '' },
  ]
},
{
  number: 18,
  name: 'Fresh and Waste Water',
  items: [
    { number: '18.0', description: 'Fresh water system', notes: '' },
    { number: '18.1', description: 'Heads / toilets', notes: '' },
    { number: '18.2', description: 'Holding tanks', notes: '' },
    { number: '18.3', description: 'Watermaker', notes: '' },
  ]
},
{
  number: 19,
  name: 'Electrical System and Generators',
  items: [
    { number: '19.0', description: 'Shore power connection', notes: '' },
    { number: '19.1', description: 'DC battery bank', notes: '' },
  ]
}

```

```

    { number: '19.2', description: 'Switchboards and distribution', notes: '' }
    { number: '19.3', description: 'Generator(s) condition', notes: '' },
    { number: '19.4', description: 'Generator exhaust and cooling', notes: '' }
    { number: '19.5', description: 'Solar / wind generation', notes: '' },
  ]
},
{
  number: 20,
  name: 'Fire and Safety Equipment',
  items: [
    { number: '20.0', description: 'Fixed fire suppression (engine room)', note
    { number: '20.1', description: 'Portable extinguishers', notes: '' },
    { number: '20.2', description: 'Life rafts', notes: '' },
    { number: '20.3', description: 'Flares and pyrotechnics', notes: '' },
    { number: '20.4', description: 'Life vests / PFDs', notes: '' },
    { number: '20.5', description: 'EPIRB', notes: '' },
    { number: '20.6', description: 'MOB equipment', notes: '' },
  ]
},
{
  number: 21,
  name: 'Navigation Equipment',
  items: [
    { number: '21.0', description: 'Chartplotters / MFD', notes: '' },
    { number: '21.1', description: 'Autopilot', notes: '' },
    { number: '21.2', description: 'Depth sounder', notes: '' },
    { number: '21.3', description: 'VHF radio', notes: '' },
    { number: '21.4', description: 'Navigation lights', notes: '' },
    { number: '21.5', description: 'AIS transponder', notes: '' },
    { number: '21.6', description: 'Radar', notes: '' },
    { number: '21.7', description: 'Satellite / internet comms', notes: '' },
  ]
},
{
  number: 22,
  name: 'Miscellaneous Equipment',
  items: [
    { number: '22.0', description: 'Stabilisers', notes: '' },
    { number: '22.1', description: 'Tenders and water toys', notes: '' },
    { number: '22.2', description: 'Davits and crane', notes: '' },
    { number: '22.3', description: 'Ships horn', notes: '' },
    { number: '22.4', description: 'Other equipment', notes: '' },
  ]
},
{
  number: 23,
  name: 'Comments and Recommendations',

```

```
        isAutoGenerated: true,
        // Auto-collects all items where recommendation_level is set
    },
    {
        number: 24,
        name: 'Surveyors Declaration',
        isDeclaration: true,
        // Shows standard declaration text + surveyor signature
    },
    {
        number: 25,
        name: 'Pictures',
        isPictureGallery: true,
        // Shows all photos from all sections in order
    },
    {
        number: 26,
        name: 'Sea Trial',
        isSeaTrial: true,
        // Special screen for sea trial data entry
    },
];
```

SEA TRIAL SCREEN

SEA TRIAL

DATE	[ 28 Nov 2025]
LOCATION	[Antibes_____]
WEATHER	[Sunny and calm____]
SEA STATE	[Calm_____]

NARRATIVE

[The engines started easily, with no
excessive smoke or vibration noted...]

ENGINE DATA TABLE

	1000	1500	2000	MAX
Coolant P	[]	[]	[]	[]
Coolant S	[]	[]	[]	[]
Oil Prs P	[]	[]	[]	[]
Oil Prs S	[]	[]	[]	[]

Speed Kts [] [] [] []

[+ Add RPM column]

TICKOVER

RPM: [600] Speed: [3.1] kts

ADDITIONAL OBSERVATIONS

[The lack of antifouling was responsible...]

PHOTO CAPTURE (in-context)

When surveyor taps  Add] in any item:

Add photo to item 8.1

 Camera]  Library]

Max 5 photos per item

Photos will be numbered automatically (pic001, pic002...)

Photos are:

- Compressed before upload (same as My Yacht photos)
 - Uploaded to Supabase Storage under survey/{report_id}/
 - Referenced as pic001 through pic999 in auto-numbering
 - Shown in section context AND collected in Section 25 (Pictures)
-

RECOMMENDATIONS AUTO-COLLECTION

Section 23 is auto-generated — never manually filled.

```
function collectRecommendations(items) {
  const recs = { A: [], B: [], C: [], D: [] };

  for (const item of items) {
    if (item.recommendation_level && item.recommendation_text) {
      recs[item.recommendation_level].push({
```

```

        section: item.section_name,
        itemNumber: item.item_number,
        text: item.recommendation_text,
    });
}
}

return recs;
}

```

In PDF, recommendations appear grouped by level:

RECOMMENDATIONS

Recommendation 'A' – Urgent:

Draw the drive shafts and replace the two PSS units at next haul out.

Free the engine raw water seacock valves.

Reseal the two fuel tank inspection plates.

...

Recommendation 'B':

Lift vessel and pressure wash hull, rudders and stern gear.

Service the stabiliser system.

Recommendation 'C':

Repair the cracked aft doorway.

Replace damaged portlight board.

PDF GENERATION

Generate a professional PDF matching the reference report format.

Page structure:

Page 1 — Cover:

- Surveyor logo (top left)
- "REPORT OF SURVEY" heading
- Vessel photo (if available)
- Details table: NAME / REGISTRATION / YEAR / MANUFACTURER / MODEL / LYING / HIN / DATE

- Surveyor association logos (YDSA / IIMS / SCMS etc.) — if uploaded

Pages 2-N — Content sections:

- Section number and title as header
- Numbered items with condition, notes, recommendations
- Recommendations in blue italic per reference format
- Photo references inline (pic001, pic002...)
- Page number bottom right + surveyor monogram

Recommendations summary page

Declaration page:

- Declaration text
- Surveyor name, qualification
- Signature image

Pictures section:

- All photos in numbered grid (3 per row)
- Caption: section name + item number

Sea Trial page:

- Header with NAME / DATE / LOCATION / WEATHER
- Narrative text
- RPM data table

CSS rules (critical):

```
@page { size: A4; margin: 15mm 20mm; }
body { font-family: 'Helvetica Neue', Helvetica, Arial, sans-serif; font-size: 9.
-webkit-print-color-adjust: exact;
print-color-adjust: exact;

/* Recommendations in blue italic */
.recommendation {
    color: #1a3a6b;
    font-style: italic;
    margin-top: 6px;
```



```

}
.rec-level-a { font-weight: 700; }

/* Section headers */
.section-header {
  font-size: 11pt;
  font-weight: 700;
  border-bottom: 2px solid #1a1a1a;
  padding-bottom: 4px;
  margin-bottom: 12px;
}

```

SURVEYOR PROFILE (in Profile tab)

Add a "Surveyor Profile" section for users on Surveyor plan:

SURVEYOR PROFILE

Name: Captain Henry Topham
 Qualification: Dip.MS, YDSA Member
 Company: Marine Developments Int.
 Phone: +34 664 890 740
 Logo: [uploaded]
 Signature: [drawn]

This data auto-fills every new survey report.

SIGNATURE PAD COMPONENT

```

// SignaturePad.jsx
// Uses react-native-signature-canvas
// Surveyor draws signature with finger
// Saved as PNG to Supabase Storage
// Used on cover page of PDF

import SignatureCanvas from 'react-native-signature-canvas';

// Props: onSave(signatureDataURL)
// Shows: white canvas with "Sign here" placeholder
// Buttons: [Clear] [Save signature]

```

FILE STRUCTURE

```
/screens/SurveyBuilder/  
  SurveyListScreen.js      ← list of reports  
  SurveySetupScreen.js     ← cover info input  
  SurveySectionsScreen.js  ← 26 sections list  
  SurveySectionScreen.js   ← section items  
  SurveyItemScreen.js      ← single item detail  
  SeaTrialScreen.js        ← sea trial data  
  SurveyPreviewScreen.js   ← PDF preview  
  
/components/SurveyBuilder/  
  SectionCard.js           ← section in list  
  SurveyItemRow.js         ← item in section  
  ConditionPicker.js       ← condition bottom sheet  
  RecommendationPicker.js   ← rec level picker  
  SignaturePad.js          ← signature drawing  
  MoistureInput.js         ← moisture reading input  
  
/utils/SurveyBuilder/  
  surveyTemplates.js       ← 26 section templates  
  recommendationCollector.js ← auto-collect all recs  
  surveyPDFBuilder.js      ← HTML → PDF generator  
  surveyPhotoManager.js    ← photo upload + numbering  
  
/hooks/  
  useSurveyReport.js       ← CRUD for survey_reports  
  useSurveyItems.js        ← CRUD for survey_items  
  useSeaTrial.js           ← CRUD for survey_sea_trial
```

PRIORITY BUILD ORDER

1. Supabase tables + RLS
2. `surveyTemplates.js` — all 26 section defaults
3. `SurveySetupScreen.js` — cover info form
4. `SurveySectionsScreen.js` — sections list with progress
5. `SurveySectionScreen.js` — section items view
6. `SurveyItemRow.js` + `ConditionPicker.js` + `RecommendationPicker.js`

7. Photo capture per item
 8. `recommendationCollector.js` — auto Section 23
 9. `SeaTrialScreen.js` — sea trial data entry
 10. `SignaturePad.js` — surveyor signature
 11. `SurveyListScreen.js` — list + draft/complete status
 12. `surveyPDFBuilder.js` — PDF generation
 13. Surveyor Profile in Profile tab
 14. Link to My Yacht passport (future)
-

MONETISATION

- Surveyor plan: \$89/month
- Free plan: NOT available for this tool
- If non-Surveyor user taps: show upgrade prompt

This tool is for professional surveyors.

Upgrade to Surveyor Plan to access:

- Full survey report builder
- Unlimited reports
- PDF export with your branding
- Signature and logo on reports

\$89 / month

[Upgrade to Surveyor →]

IMPORTANT NOTES

1. Reports auto-save every 30 seconds — surveyor never loses work
2. Works offline — saves to local AsyncStorage, syncs when online
3. Photos numbered automatically across entire report (pic001, pic002...)
4. Section 3 (Glossary) and Section 24 (Declaration) have standard text pre-filled

5. Section 23 (Recommendations) is read-only — auto-populated from all items
 6. Section 25 (Pictures) shows all photos from all sections in order
 7. Surveyor can add custom items to any section (tap + Add item)
 8. Items can be reordered by long-press drag
 9. Report can be exported as PDF at any stage (draft is OK)
 10. PDF uses A4 with 15mm margins — NOT full-bleed like proposal template
 11. Recommendations in PDF are blue italic — matches industry standard format
 12. Each PDF page shows surveyor monogram (initials) bottom right
-

YachtWorth — Survey Report Builder Target: Professional marine surveyors · Surveyor plan \$89/mo Reference: YDSA / IIMS / SCMS survey report format 26 sections · Auto recommendations · PDF with logo + signature